AOU
الجامعة العربية المفتوحة
Arab Open University
Faculty of Computer Studies

TM352
Web, Mobile and Cloud Technologies

Mid-term Examination MTA - Makeup
Fall Semester 2025/2026

Date: 3/Dec. /2025

Number of Exam Pages: 3 (including this cover sheet)	**Time Allowed: 2 Hrs**

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Instructions:

- Total Marks: 45 {will be adjusted to 30 marks}
- This exam consists of 3 parts.
- **ALL questions** must be answered in the External Answer booklet.
- Be sure you write your **name and ID** on the External Answer booklet.
- **Calculators** are not allowed.

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PART I (5 Marks)

Answer all the questions

Choose the right answer

1. Which of the following best describes the purpose of a *digital signature*?
A) To encrypt the contents of a message for privacy
B) To verify the author of a message and ensure it hasn't been altered
C) To establish a shared secret key between two parties
D) To compress data before transmission

2. _____ assures that information and programs are changed only in a specified and authorized manner
A) System Integrity
B) Program Integrity
C) Data Integrity
D) Data Privacy

3. Which protocol is responsible for reliably delivering data in order between two endpoints on the internet?
A. UDP
B. FTP
C. TCP
D. IP

4. Which of the following HTTP method submit data to the server
A) GET
B) POST
C) HEAD
D) PUT

5. In the Diffie–Hellman key exchange protocol, which of the following statements is TRUE?
A) Secret keys must be exchanged physically or through a courier
B) The same secret key is directly transmitted over the network
C) Both parties independently compute a shared secret without transmitting it
D) Public and private key pairs are not required for the exchange to work

PART II (20 Marks)

Answer all the questions

1. Describe how *digital signatures* ensure message authenticity and integrity. Discuss the role of public and private keys in this process. [5 Marks]

2. Explain the difference between privacy, authenticity, and integrity in web security. Provide an example of how each concept is applied in the exchange of sensitive data between a client and a server. [5 Marks]

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3. Alice and Bob are using the **Diffie–Hellman key exchange** method to establish a shared secret key. They agree on a common plaintext value x=4. Alice chooses her secret key a=2, and Bob chooses his secret key b=3. The encryption operation is exponentiation.

Show the sequence of calculations each party performs and determine the **shared secret key** they both end up with. [10 Marks]

PART III (20 Marks)

Answer all the questions

1. When typing in the address part of the web browser
./webresources/generic/convert? name=**

Complete the following RESTful web service, by writing the service annotations [5 marks]

```
@Path("----------")
public class GenericResource {
@-----
@Path("--------")
@Produces(MediaType.TEXT_HTML)
public String modify(@-------Param("name") String name) {
return name.toUpper();}
}
```

2. Discuss three main components of Service-Oriented Architecture(SOA). [4 Marks]

3. Design a simple login form redirecting to "login.jsp" with post method using HTML5 with input fields for email and password, and include a Submit button. Use appropriate input types for each field. [5 Marks]

4. Provide XML representation for data in the following table: [6 Marks]

CourseCode	CourseName	TutorID	TutorName	SessionNumber
TM352	Cloud Tech.	T105	Mohammed	1301
TM252	Internet Programming	T103	Ahmed	3201
TM351	Data Analysis	T502	Khaled	5102
M251	Java Programming	T105	Mohammed	5101

Table 1: Course-Tutor-Data

End of questions

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